Pronob Kumar Barman

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EDUCATION

University of Maryland, Baltimore County (UMBC) — Expected Dec 2025
Ph.D. Information Systems, GPA: 3.85/4.00
Research: Generative Models, NLP, Recommendation Systems, HCI

University of Kentucky — May 2021
M.S. Statistics, GPA: 3.85/4.00

University of Dhaka — Jan 2017
M.S. Statistics, GPA: 3.46/4.00
B.S. Statistics, GPA: 3.46/4.00

PROFESSIONAL EXPERIENCE

PhD Researcher — UMBC — Jan 2022–Present

- Led human-centered AI research at the intersection of healthcare and machine learning, designing fair and interpretable models to improve patient outcomes in online health communities.
- Spearheaded development of **Generative Peer Support Recommender**, a novel framework integrating machine learning with social science theories to automate personalized support group formation. Key technical contributions:
  - Created **Group-specific Dirichlet Multinomial Regression (gDMR)**: Extended DMR models with group-specific parameters using BERT embeddings, user demographics, and interaction patterns (Python).
  - Engineered **Group-specific Structural Topic Model (gSTM)**: Enhanced STM stability by 60% through group-aware topic modeling and network embeddings (R).
- Achieved **75% higher recommendation accuracy** vs. baseline models in real-world experiments, measured via A/B testing with patient engagement metrics (e.g., post frequency, community retention).
- Open-sourced implementations as widely adopted tools: Published gSTM R package and gDMR Python Code, enabling scalable deployment in healthcare AI research.

Teaching Assistant — UMBC — Jan 2022–Present

- Designed and conducted workshops on **Machine Learning (ML)**, **LLM workflows**, and system reliability for real-world applications.
- Assisted students in deploying scalable ML pipelines and models on **cloud platforms** like AWS and Vertex AI.

Machine Learning Engineer — Technuf LLC — Jun 2024–Aug 2024

- Developed **retrieval-augmented generation (RAG)** pipelines for cybersecurity, integrating LLMs such as **BERT** and **GPT models** for advanced text retrieval and reasoning tasks.
- Conducted advanced trend analysis on alert data, reducing **false positives by 20%**, and **synthesized actionable insights** for business impact.

Machine Learning Engineer — Technuf LLC — Aug 2022–Dec 2022

- Streamlined data processing for **200K+ healthcare records** using optimized **SQL queries** in PostgreSQL, improving system efficiency by **40%**.
- Built interactive **Power BI dashboards** to visualize critical healthcare metrics, enabling **data-driven decision-making**.
- Automated testing workflows with **Flutter**, reducing debugging time and improving product release efficiency.

Deputy Director, Data Science — Bangladesh Bank — Apr 2018–Dec 2021

- Developed predictive credit risk models, reducing defaults by 10%.
- Enhanced foreign investment strategy analytics, increasing FDI inflows by 15%.

TECHNICAL SKILLS

- Languages: Python, R, Java, SAS, SQL, MATLAB
- ML Frameworks: PyTorch, TensorFlow, scikit-learn
- MLOps: Docker, Kubernetes, AWS, GCP
- NLP: BERT, GPT, SpaCy, NLTK
- Data Viz: Tableau, Power BI, Matplotlib
- Data Management: PostgreSQL, Pinecone, ChromaDB
- Generative AI: GPT-3/4, LangChain, Hugging Face
- Research: NVivo, Qualtrics, LaTeX, VScode, Git

SELECTED PROJECTS

DistilBERT-based Question Answering System — GitHub

- Deployed **DistilBERT** models on AWS SageMaker for real-time customer **query resolution**.
- Reduced response time by **30%** through scalable API integration.

Deploying a Machine Learning Model for Digit Classification on Google Cloud Run — Github

- Developed and deployed a scalable **Flask-based web** app using **TensorFlow**, containerized with **Docker**, and hosted on **Google Cloud Run** to enable real-time predictions through a public API endpoint.

Air Quality Prediction and Deployment on Google Cloud Platform Github

- Designed and deployed a scalable **Air Quality Index (AQI)** forecasting system using linear regression, containerized with **Docker**, and integrated with a **Flask** web app on **Google Cloud Platform** for interactive user predictions.

LLM Generative AI with LangChain and OpenAI API Github

- Developed a **retrieval-augmented generation system** for advanced semantic search, using **Pinecone and ChromaDB**.

End-to-End Medical Chatbot with Llama2 and Hugging Face API Github

- Developed a **medical chatbot with Llama2**, improving **patient interaction flow by 30%** through **LangChain** enhancements.

Cognitive and Affective State Recognition in iVR Using Multimodal Signals Github

- Developed inverse inference models using **multimodal signals**, integrating neuro-physiological data such as EEG and eye-tracking to map cognitive and affective states in immersive virtual reality (iVR).

Predictive Policing with GAN

- Implemented a **Generative Adversarial Network (GAN)** model to identify and mitigate bias in predictive policing algorithms, achieving a 20% reduction in false-positive rates.

PUBLICATIONS

- **Barman, P. K.**, Reynolds, T. L., & Foulds, J. (2025). "Facilitating Online Healthcare Support Group Formation using Topic Modeling." *Proceedings of the 19th World Congress on Medical and Health Informatics (MedInfo 2025)*, 2025.
- Rachael, K., Reynolds, T. L., Yus R. & **Barman, P. K.** (2024). "Acceptance of Occupancy Dashboards among Undergraduate Students: Towards Everyday Decision-making Infrastructure that can be Leveraged in Pandemics." *ACM Conference on Computer-Supported Cooperative Work and Social Computing*, 2024. (Under Review)
- Walid, M. A. A., Devnath, M. K., Muiz, B., Melon, M. M. H., **Barman, P. K.**, & Barman, P. K. (2024). "Efficient Graph-Based Intrusion Detection for CAN Bus Security with Minimal Training Data." *Proceedings of the 6th IEEE International Conference on Sustainable Technologies for Industry 5.0 (STI 2024)*, 2024.
- Balogun, Z. A., **Barman, P. K.**, Onwumbiko, B. K., & Reynolds, T. L. (2024). "User-generated Data for Different Types of Mobile-Personal Health Records: A Computationally-guided Qualitative Analysis Approach." *AMIA Annual Symposium, 2024*.
- Faruq, M. O., Walid, M. A. A., Baowaly, M. K., Devnath, M. K., Ejaz, M. S., **Barman, P. K.**, & Sattar, A. S. (2024). "Shielding Privacy: A Technique of Extenuating Composition Attacks in Various Independent Data Publication." *Bulletin of Electrical Engineering and Informatics, 13(4), 2677-2686*.
- **Barman, P. K.** (2023). "Detecting Abnormal Health Conditions in Smart Home Using a Drone." *arXiv preprint arXiv:2310.05012*.

GRANTS & AWARDS

UMBC GSA Research Grant (2024): $1,000 for research on healthcare group formation models.

PEER REVIEWER

- NeurIPS 2024 Workshop on Behavioral ML
- AMIA 2024 Annual Symposium